

EPGSpeller: Open-Vocabulary Silent Spelling with Electropalatography

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Abstract

We study lexicon-free (character-level) silent spelling from electropalatography (EPG) under a word-holdout protocol and ask whether spatial inductive biases derived from a fixed electrode-to-grid proxy layout improve character-level decoding. We evaluate three EPG front-end representations under matched training conditions: (i) a layout-agnostic channel vector encoder, (ii) a layout-aware two-dimensional reconstruction consumed by convolution, and (iii) a layout-aware row/column compression. In this setting, the layout-aware two-dimensional convolutional front-end does not outperform the vector baseline, while spatial augmentation improves robustness to fixed electrode dropouts.

(d21d93c9ec3c). A channel is *mapped* if $\pi(c)$ is defined (120) and *unmapped* otherwise (4). When multiple mapped channels share a grid cell (5channels), we aggregate them deterministically by averaging to construct each grid frame and include a separate binary mask channel indicating electrode presence. This proxy mapping is used only as an inductive bias for layout-aware representations; we do not interpret it as a geometric tongue model.

We use a character-level connectionist temporal classification (CTC) formulation [7]. In the word-holdout protocol, training and test sets are disjoint at the word-identity level: all examples of each test word are excluded from training. As a result, greedy CTC decoding is not constrained to a fixed training lexicon.

1 Problem setting

Electropalatography (EPG) measures tongue–palate contact as a multichannel time series from a fixed electrode array. EPG has long been used to study and visualize tongue–palate contact patterns in phonetic research and speech training (e.g., [8, 9, 14]), and prior work has explored EPG data reduction and representation [10, 1]. We work with a fixed inventory of 124electrode channels indexed by the SmartPalate channel numbering, and we use that index order as the canonical channel order for layout-agnostic models.

To introduce spatial structure, we define a fixed proxy mapping π from SmartPalate channel indices to grid coordinates on a 16×16 grid, derived from the pinned SmartPalate distribution snapshot `ev_scripts_smartpalate_distribution_csv`

2 Related work and positioning

Silent speech interfaces (SSI) aim to infer linguistic content without relying on the acoustic speech signal [2, 6, 4]. SSI has been explored with multiple sensing modalities, including ultrasound and optical imaging of the tongue and lips [11] and implanted magnetic-sensor setups [5]; in this work we focus specifically on electropalatography (EPG). Within EPG-based SSI, silent spelling and text entry have been explored previously (e.g., [12]), and silent spelling has also been studied with other wearable sensors [3]. Our contribution is a controlled EPG representation study: under a fixed word-holdout protocol and a fixed CTC decoder, we isolate the effect of layout-agnostic (vector) versus layout-aware (proxy-grid two-dimensional and row/column) front-ends, and we analyze robustness to fixed electrode dropouts via spatial augmen-

tation. We therefore scope claims to this protocol and proxy-layout assumption, and do not claim new hardware, a new electrode geometry, or a new decoding objective.

3 Models and spatial front-ends

All variants share the same temporal backbone and CTC decoding setup. The temporal backbone produces 64-dimensional per-frame embeddings and uses the same recurrent stack across conditions (uni-gru; 5layers; 512hidden units). We apply a temporal unfolding operator (sliding-window chunking) with kernel length 32 and stride 4 before the recurrent stack; we vary only the EPG spatial front-end described below.

Vector (layout-agnostic). The vector baseline uses the canonical channel index order and does not use any electrode layout information.

Layout-aware two-dimensional reconstruction. We scatter the selected channels into the 16×16 proxy grid and apply a two-dimensional convolutional front-end to obtain per-frame embeddings. Grid cells without electrodes are masked, duplicated channels are aggregated deterministically, and unmapped channels are embedded separately and combined with the grid embedding.

Row/column compression. We form row-wise and column-wise pooled summaries from the same proxy grid, concatenate them (optionally with the unmapped channels), and project to the shared embedding size.

4 Experimental protocol

We evaluate the word-holdout protocol using 4 random split seeds (seed0, seed1, seed2, seed3) and reuse the exact same splits across all compared spatial front-ends. Training uses 10000 optimization steps at 10.0 ms frame shift and enables SpecAugment-inspired masking [13].

We compare two electrode-selection baselines for budgeted subsets of size K : *TopK*, which selects the K highest-ranked channels under an occlusion-derived importance score, and *FPS-diverse TopK*, which augments

Table 1: Vector vs two-dimensional convolution (mean $\pm$ std over splits). Evidence: `ev_ed20260217_h11_spatial2d_vs_vector_pl_k32_k64_k5(2f2840f6dee5)`.

K	Vector Greedy CER	Spatial2D Greedy CER	Vector Stream RTF	Spatial2D Stream RTF
TopK				
32	0.1399 $\pm$ 0.0162	0.4885 $\pm$ 0.1269	0.000491 $\pm$ 0.000035	0.002670 $\pm$ 0.000127
64	0.1289 $\pm$ 0.0135	0.2609 $\pm$ 0.0300	0.000476 $\pm$ 0.000004	0.002895 $\pm$ 0.000304
96	0.1194 $\pm$ 0.0173	0.2304 $\pm$ 0.0350	0.000475 $\pm$ 0.000023	0.002767 $\pm$ 0.000238
124	0.1048 $\pm$ 0.0093	0.1831 $\pm$ 0.0103	0.000469 $\pm$ 0.000014	0.002639 $\pm$ 0.000067
FPS2K				
32	0.1440 $\pm$ 0.0136	0.5795 $\pm$ 0.3052	0.000514 $\pm$ 0.000022	0.002674 $\pm$ 0.000129
64	0.1195 $\pm$ 0.0103	0.3581 $\pm$ 0.3079	0.000532 $\pm$ 0.000094	0.002808 $\pm$ 0.000180
96	0.1137 $\pm$ 0.0161	0.3438 $\pm$ 0.3104	0.000486 $\pm$ 0.000029	0.002626 $\pm$ 0.000096
124	0.1048 $\pm$ 0.0093	0.1831 $\pm$ 0.0103	0.000469 $\pm$ 0.000014	0.002639 $\pm$ 0.000067

the TopK ranking with a proxy-grid spatial-diversity constraint (farthest-point sampling from a bounded candidate pool). We treat both as baselines and do not claim novelty for these heuristics.

We report character error rate (CER) from greedy CTC decoding and streaming real-time factor (RTF). CER is computed as total edit distance divided by total reference characters on the evaluation set. RTF is computed as total model forward-pass wall time divided by total input duration (derived from the number of input frames and the 10.0 ms frame shift). Lower is better for both metrics.

5 Results

Scope. All comparisons below use the same protocol and matched training settings; we interpret the findings as evidence about spatial inductive biases under this setting.

5.1 Two-dimensional convolution vs vector

Table 1 compares vector and two-dimensional convolutional front-ends for multiple electrode budgets. In this protocol and setting, the layout-aware two-dimensional convolutional front-end is worse than the vector baseline in greedy CER and also slower in streaming RTF.

5.2 Row/column compression vs vector

Table 2 shows that row/column compression stays close to the vector baseline in greedy CER, but increases streaming RTF.

Table 2: Vector vs row/column compression (mean $\pm$ std over splits). Evidence: ev_ed20260217_h13_rowcol_vs_vector_pl_k32_k64_2026-02-17 (a7fb56ec370d).

K	Vector Greedy CER	RowCol Greedy CER	Vector Stream RTF	RowCol Stream RTF
TopK				
32	0.1399 $\pm$ 0.0162	0.1448 $\pm$ 0.0204	0.000491 $\pm$ 0.000035	0.002141 $\pm$ 0.000276
64	0.1289 $\pm$ 0.0135	0.1269 $\pm$ 0.0143	0.000476 $\pm$ 0.000004	0.002072 $\pm$ 0.000142
FPS2K				
32	0.1440 $\pm$ 0.0136	0.1466 $\pm$ 0.0080	0.000514 $\pm$ 0.000022	0.002176 $\pm$ 0.000311
64	0.1195 $\pm$ 0.0103	0.1255 $\pm$ 0.0193	0.000532 $\pm$ 0.000094	0.002104 $\pm$ 0.000216

Table 3: Fixed-dropout robustness for the two-dimensional front-end (mean $\pm$ std over splits). Values are Δ CER relative to clean evaluation. Evidence: ev_ed20260217_h12_spatial_aug_fixed_dropout_summary_md-topk164_2026-02-17 (0cd1fecedc16).

Drop rate	No spatial aug	Spatial aug
0.0	0.0000 $\pm$ 0.0000	0.0000 $\pm$ 0.0000
0.1	0.0245 $\pm$ 0.0027	0.0183 $\pm$ 0.0065
0.2	0.0783 $\pm$ 0.0162	0.0660 $\pm$ 0.0097
0.3	0.1440 $\pm$ 0.0183	0.1224 $\pm$ 0.0120

5.3 Spatial augmentation and robustness to electrode failures

To quantify robustness, we apply fixed electrode-dropout masks at evaluation time and report the increase in greedy CER relative to clean evaluation. Here, *spatial augmentation* refers to training-time masking or perturbations that operate on the proxy electrode grid (rather than treating channels as exchangeable). Table 3 shows that spatial augmentation reduces the CER degradation under dropout.

6 Artifacts and auditability

This project maintains a claim-evidence registry in `paper/paper.json`. All numeric facts and result tables in this draft are generated from pinned evidence snapshots via `scripts/paper/generate_facts_tex.py` using `paper/facts_spec.yaml`. A strict checker rejects handwritten numeric facts in the manuscript

body. To regenerate the manuscript numbers from a clean checkout, run: `python3scripts/paper/generate_facts_tex.py--facts_specpaper/facts_spec.yaml--paper_jsonpaper/paper.json--root.` and `python3scripts/paper/check_no_manual_numeric_facts.py--texmain.tex--facts_specpaper/facts_spec.yaml--root..` To audit claim-evidence consistency, run: `python3tools/paper_jsonLint.py--paper-jsonpaper/paper.json--root.--strict` and `python3scripts/render_paper_json_claim_audit.py--paper-jsonpaper/paper.json--root.--out-mdpaper/snippets/claim_audit_md-topk164_2026-02-17`. Key pinned evidence identifiers used by this draft include `ev_scripts_smartpalate_distribution_csv(d21d93c9ec3c)`, `ev_ed20260217_h11_spatial2d_vs_vector(2f2840f6dee5)`, `ev_ed20260217_h13_rowcol_vs_vector_pl(a7fb56ec370d)`, and `ev_ed20260217_h12_spatial_aug_fixed_dropout_summary(0cd1fecedc16)`.

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